East West University
Department of CSE

Course Code: **CSE 347**
Course Title: **Information System Analysis and Design**

Project: **StudyByte – Peer-to-Peer Micro-Tutoring**

Task: **CPM & PERT Analysis**

Semester and Year:
Summer 2025

GROUP NO: **04**

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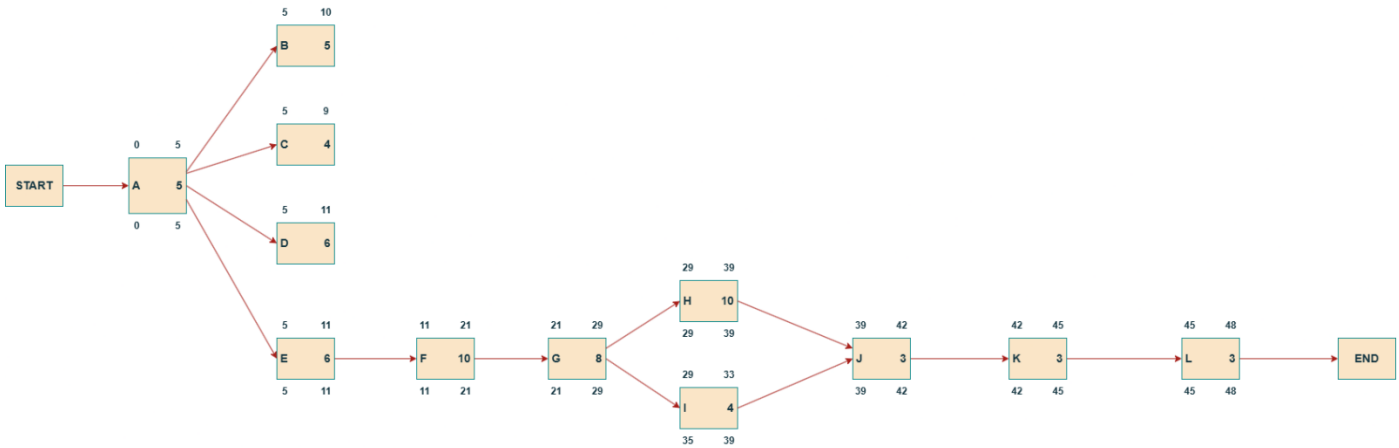
01/04/2026

CPM & PERT Analysis

Part 1: Critical Path Method (CPM)

1.1 CPM Network Diagram



1.1 CPM Table

ES = Early Start | **EF** = Early Finish

LS = Late Start | **LF** = Late Finish

TF = Total Float | **FF** = Free Float

CP = Critical Path

Activity	ID	Dependency	Duration	ES	EF	LS	LF	Total Float	Free Float	CP ✓
Requirement Analysis	**A**	-	**5**	0	5	0	5	**0**	**0**	✓
Feasibility Analysis	B	A	5	5	10					
Front-End Design	C	A	4	5	9					
System Design	D	A	6	5	11					
Database Design	**E**	A	**6**	5	11	5	11	**0**	**0**	✓
Implementing User Module Features	**F**	A,E	**10**	11	21	11	21	**0**	**0**	✓
Implementing Booking Module	**G**	A,F	**8**	21	29	21	29	**0**	**0**	✓
Implementing Wallet & Escrow	**H**	E,F,G	**10**	29	39	29	39	**0**	**0**	✓
Implementing Rating Feature	I	F,G	4	29	33	35	39	6	6	

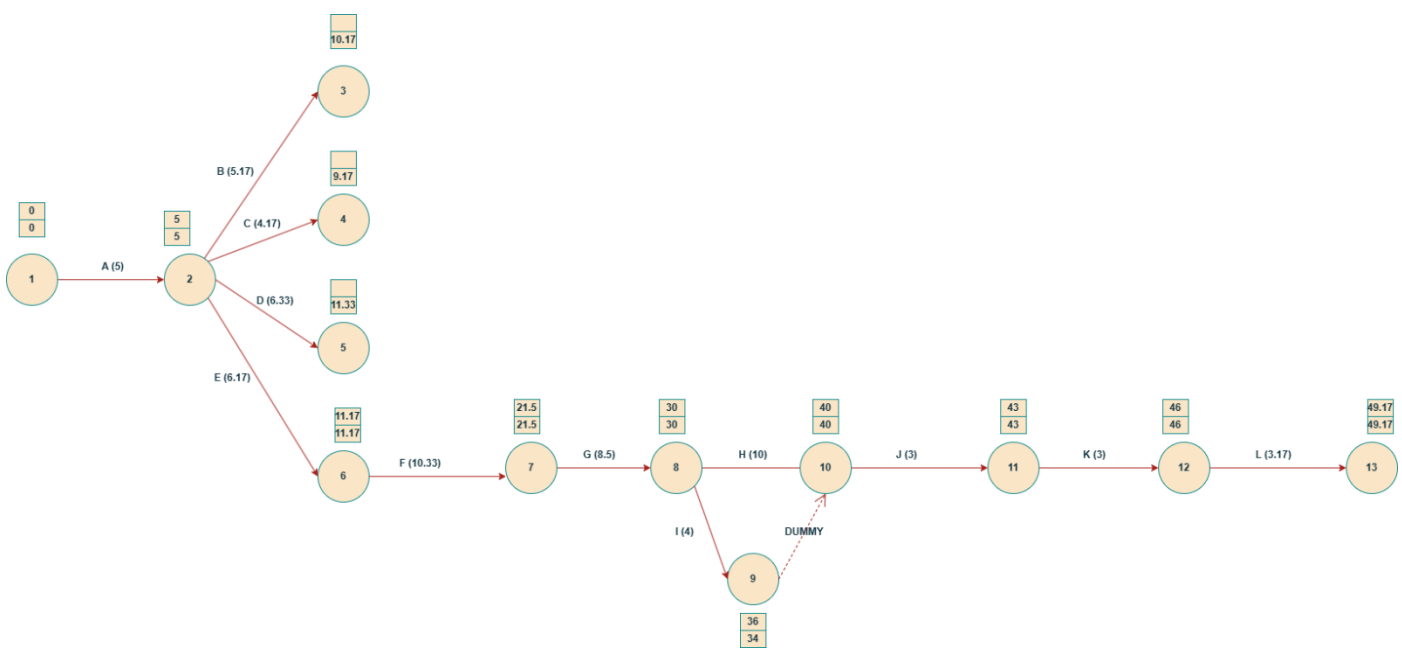
Activity	ID	Dependency	Duration	ES	EF	LS	LF	Total Float	Free Float	CP ✓
Implementing Admin Control	**J**	H,I	**3**	39	42	39	42	**0**	**0**	✓
Testing	**K**	A,F,G,H,I,J	**3**	42	45	42	45	**0**	**0**	✓
Documentation	**L**	K	**3**	45	48	45	48	**0**	**0**	✓

1.2 CPM Summary

Metric	Value
Project Length (Duration)	**48 Days**
Critical Path	**A → E → F → G → H → J → K → L**
Number of Critical Activities	**8 out of 12**
Non-Critical Activities	B, C, D, I

Part 2: Program Evaluation & Review Technique (PERT)

2.1 PERT Network Diagram



2.2 PERT Table

O = Optimistic | **M** = Most Likely (original) | **P** = Pessimistic

te = Expected Time = (O + 4M + P) / 6

σ² = Variance = $((P-O)/6)^2$

Activity	ID	Dependency	O (days)	M (days)	P (days)	te (days)	σ^2 (Var)	CP ✓
Requirement Analysis	**A**	-	3	5	7	**5**	0.44	✓
Feasibility Analysis	B	A	3	5	8	5.17	0.69	
Front-End Design	C	A	3	4	6	4.17	0.25	
System Design	D	A	4	6	10	6.33	1	
Database Design	**E**	A	4	6	9	**6.17**	0.69	✓
Implementing User Module Features	**F**	A,E	8	10	14	**10.33**	1	✓
Implementing Booking Module	**G**	A,F	7	8	12	**8.5**	0.69	✓
Implementing Wallet & Escrow	**H**	E,F,G	6	10	14	**10**	1.78	✓
Implementing Rating Feature	I	F,G	2	4	6	4	0.44	
Implementing Admin Control	**J**	H,I	2	3	4	**3**	0.11	✓
Testing	**K**	A,F,G,H,I,J	1	3	5	**3**	0.44	✓
Documentation	**L**	K	2	3	5	**3.17**	0.25	✓

2.2 PERT Summary

Metric	Value
PERT Critical Path	**A → E → F → G → H → J → K → L**
Expected Project Length (ΣTE)	**49.17 Days**
Sum of Variances (σ^2) on Critical Path	5.40 days^2
Standard Deviation (σ)	2.32 days